

```
simplebot.py  x
1  # WRITE YOUR CODE HERE
2  import tkinter.scrolledtext as tks #creates a scrollable text window
3
4  from datetime import datetime
5  from tkinter import *
6
7  # Create the main application window using Tk()
8  baseWindow = Tk()
9
10 # Set the title of the window
11 baseWindow.title("The Simple Bot")
12
13 # Set the size of the window
14 baseWindow.geometry("500x250")
15
16 # Create the chat window as a ScrolledText widget with "Arial" font
17 chatWindow = tks.ScrolledText(baseWindow, font="Arial")
18
19 # Configure tags for message alignment: 'tag-left' for bot messages, 'tag-right' for user messages
20 chatWindow.tag_configure('tag-left', justify='left')
21 chatWindow.tag_configure('tag-right', justify='right')
22
23 # Disable the chat window initially (it should not be editable by the user)
24 chatWindow.config(state=DISABLED)
25
26 # Create the send button, with specific font, text, and background color
27 # The 'command' option is commented out. Uncomment it and replace 'send' with your send function's name
28 sendButton = Button(
29     baseWindow,
30     font=("Verdana", 12, 'bold'),
31     text="Send",
32     bg="#fd94b4",
33     activebackground="#ff467e",
34     fg='ffffff',
35     # command=send
36 )
37
38 # Create the user entry box where the user types their messages
39 userEntryBox = Text(baseWindow, bd=1, bg="white", width=38, font="Arial")
40
41 # Place the chat window, user entry box, and send button on the main window using specific coordinates and
42 # chatWindow.place(x=1, y=1, height=200, width=500)
```

Guide x

Collapse

A Simple Chatbot in Python

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Initializing the GUI

A simple, rule-based chatbot.

Writing Code

To your left, you should see an opened file called `simplebot.py`. Use this file and reference the syntax below to work on this Lab.

Let's start by building a GUI for our chatbot. We'll use ***tkinter***, the standard Python GUI library.

The first step is to import all the necessary modules into our file. Paste the following code in the top left pane:

```
import tkinter.scrolledtext as tks #creates a scrollable text window

from datetime import datetime
from tkinter import *
```

To make a simple GUI for our chatbot, it should have 4 main components:

- `baseWindow` - the main GUI window that contains everything
- `chatWindow` - displays the conversation between a user and the chatbot
- `userEntryBox` - for the user to type their queries for the Chatbot
- `sendButton` - a button that sends the user query to the Chatbot

Let's use `tkinter` to initialize the `baseWindow` and the other components and place them on the `baseWindow`.

Paste the following code in the top left pane under the import statements:

```
# Create the main application window using Tk()
baseWindow = Tk()

# Set the title of the window
baseWindow.title("The Simple Bot")

# Set the size of the window
baseWindow.geometry("500x250")

# Create the chat window as a ScrolledText widget with "Arial" font
chatWindow = tks.ScrolledText(baseWindow, font="Arial")

# Configure tags for message alignment: 'tag-left' for bot messages, 'tag-right' for user messages
chatWindow.tag_configure('tag-left', justify='left')
chatWindow.tag_configure('tag-right', justify='right')

# Disable the chat window initially (it should not be editable by the user)
chatWindow.config(state=DISABLED)

# Create the send button, with specific font, text, and background color
# The 'command' option is commented out. Uncomment it and replace 'send' with your send function's name
sendButton = Button(
    baseWindow,
    font=("Verdana", 12, 'bold'),
    text="Send",
    bg="#fd94b4",
    activebackground="#ff467e",
    fg='ffffff',
    # command=send
)

# Create the user entry box where the user types their messages
userEntryBox = Text(baseWindow, bd=1, bg="white", width=38, font="Arial")

# Place the chat window, user entry box, and send button on the main window using specific coordinates and
chatWindow.place(x=1, y=1, height=200, width=500)
userEntryBox.place(x=3, y=202, height=27)
sendButton.place(x=430, y=200)

# Start the main event loop to keep the application running and responsive
baseWindow.mainloop()
```

Look at our code, which of the following is TRUE about the `tkinter` GUI ?

- ☐ The size of the base GUI component/window cannot be changed.
- ☐ The text in a tkinter `ScrolledText` component cannot be right or left justified.
- ☒ The `.place` function is used to place a component (eg. button, textBox, etc.) on the base GUI component/window.

The text in a tkinter `ScrolledText` component can be right or left justified. This can be done by using `.tag_configure('tag-left', justify='left')` on the `ScrolledText` object.

The size of the base GUI component/window can also be changed by using the `.geometry()` function.

Check It!

Let's close the GUI and look at how we can attach a function to the send button!

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